

Participatory Design Insights for Holistic Data Collection in DLB Using Smartphones and Wearables: Toward an Adaptive Digital Assistant

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Abstract

This research explores the potential of digital sensors for holistic data collection in patients with dementia with Lewy bodies (DLB). We conducted a participatory workshop with human-computer interaction and clinical experts. This workshop explored novel data collection methods using smartphones and wearables, considering the limitations of current practices and the capabilities of available sensors. The goal was to enable holistic, multimodal data collection that captures the variety and heterogeneity of DLB symptoms. Workshop insights led to multiple ideas, including designing active tasks integrated into daily routines, prioritizing camera-based tasks due to cameras accessibility and capacity to capture a wide range of data. Findings informed the conceptualization of a digital assistant designed to adaptively personalize tasks and instructions based on users' disease progression, preferences, and previously gathered data. This adaptive framework integrates passive and active data to ensure comprehensive, user-centered monitoring of DLB symptoms, addressing key challenges identified in current methods.

CCS Concepts

- **Applied computing** → *Health informatics*; • **Human-centered computing** → **Personal digital assistants**; *User centered design*;
- **Social and professional topics** → *Assistive technologies*.

Keywords

Dementia with Lewy Bodies (DLB), participatory design, multimodal data collection, adaptive task automation, smartphone sensing, conversational AI, digital assistant, wearable sensors

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1 Introduction

Dementia with Lewy Bodies (DLB) is the second most common neurodegenerative dementia after Alzheimer's disease (AD), yet it remains under-researched despite its prevalence, associated healthcare costs, caregiver burden, and high mortality rates. DLB is frequently misdiagnosed and often confused with AD or Parkinson's disease, leading to delays in appropriate management and increased challenges for patients, caregivers, and healthcare systems. Although research in dementia has largely centered around Alzheimer's, the distinct pathological and symptomatic features of DLB require a tailored research focus. The diverse clinical representation of DLB, characterized by dementia and a varying combination of multiple symptoms such as parkinsonism, fluctuations in alertness and cognition, visual hallucinations, and sleep disorders, underscores the importance of holistic and continuous assessment [11, 15, 17, 20].

Wearable digital sensors and smartphones offer the potential to capture a broad spectrum of data encompassing diverse DLB symptoms that may be missed by traditional clinical assessments. These technologies enable continuous, longitudinal, and objective data collection, contrasting with the episodic and often subjective nature of data obtained during clinical visits. Data acquisition can occur passively through automated sensor readings or actively via

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patient interaction with smartphone-based tasks designed to elicit diagnostic information [4]. However, the adoption of digital sensors for data collection introduces methodological differences compared to traditional clinical data collection. The inherent characteristics of sensors in generating data necessitate tailored approaches to task and protocol design, ensuring alignment between the data collection strategy and the capabilities of sensors.

Effective and feasible implementation of such solutions requires the integration of diverse, domain-specific expertise throughout the design process. Clinical experts provide necessary insights into the disease's pathophysiology, patient experiences, and symptom manifestations, while specialists in Human-Computer Interaction (HCI) contribute expertise on sensor capabilities and user interaction design.

This paper contributes with a participatory co-design workshop involving clinicians and HCI specialists. The workshop aimed to develop a comprehensive protocol for longitudinal and multimodal data collection specifically tailored to the unique needs of individuals with DLB. This collaborative effort generated key themes and considerations that informed the development of an AI-powered digital assistant. Envisioned as an integrated component of patients' daily routines, this assistant serves as a central hub for multimodal data integration. Its design prioritizes enhancing patient engagement and motivation to ensure sustained participation in longitudinal data collection, while simultaneously addressing critical clinical considerations and challenges inherent in data acquisition within this population.

2 Related Work

Traditional methods of diagnosing dementia rely on episodic evaluations conducted in clinical settings. These approaches often involve time-consuming, expertise-dependent tests, such as test batteries or clinician-administrated tasks, alongside observational assessments. However, these approaches are often limited by issues such as cultural biases, practice effects, and subjectivity of tests. In addition, they fail to capture the day-to-day variability and fluctuations of symptoms, as well as longitudinal changes, which are critical for effective monitoring and diagnosis of DLB [3, 10].

Digital sensors embedded in smartphones and wearables offer a promising opportunity to address some limitations of traditional approaches. By enabling continuous and objective data collection, they facilitate the longitudinal monitoring of patients, providing insights into the variability of symptoms while offering greater precision, reproducibility, and reduced susceptibility to biases [3, 7, 10]. Wearable and smartphone sensors provide a rich source of passive and active data encompassing motor, cognitive and neuropsychiatric functions, sleep patterns, and autonomic nervous system activity, enabling comprehensive assessment of a wide range of symptoms for diagnostic and monitoring purposes. Passive data streams, often collected continuously and unobtrusively, include various gait [13] and balance [14] characteristics, physical activity levels [12, 22] (derived from IMUs), and detailed sleep pattern data [9]. Active data, typically gathered through user interaction with pre-designed tasks, includes finger tapping [22] and drawing tasks [23] via interaction with touchscreen, analysis of voice and language features [24] captured by the microphone, and assessment of facial expressions and

oculomotor data through the camera [8]. Beyond these, passive data derived from phone usage and messaging patterns—such as typing speed, call duration, conversation frequency, app usage (including task switching patterns and revisitation rates)—have been linked to cognitive and behavioral health indicators. GPS data provides some insights into mobility patterns and social engagement, which can be particularly relevant for assessing depressive symptoms [10].

Nonetheless, many existing digital approaches simply replicate traditional clinical tests without fully leveraging the unique data-gathering capabilities of sensors [1]. Furthermore, much of this work focuses on single biomarkers such as merely capturing metrics like gait, balance, or tremors, often failing to reflect the multidimensional complexity of dementia and the interplay between its diverse symptoms. To address this gap, there is growing emphasis in the literature on developing multimodal and holistic digital biomarkers, which integrate data from diverse sources to provide a deeper understanding of disease patterns and progression [7, 19, 21].

Studies have demonstrated that chatbots and conversational agents can effectively reduce social isolation and loneliness among older adults, enhancing their quality of life [2, 5]. A recent review further underscored the broader application of these tools in healthcare, where chatbots have demonstrated high acceptability and efficacy in improving the physical, psychological, and overall well-being of the elderly [25]. Additionally, another review highlighted the potential of non-invasive technologies to meet the unique needs and preferences of elderly users. Wearable devices, in particular, exemplify this trend, as older individuals often prioritize non-intrusive solutions. By integrating data collection tasks into daily routines with minimal disruption, such tools enhance usability and promote adherence. Collectively, these advancements indicate a growing emphasis on intelligent assistive technologies in dementia care, presenting opportunities for sustainable and scalable solutions that align with the needs and priorities of aging populations [6].

Co-designing with domain experts is increasingly recognized as a critical component in the development of healthcare technologies. Collaborative approaches ensure that solutions are not only technologically feasible but also practical, ethical, and aligned with the needs of end-users. For instance, integrating the expertise of clinicians, researchers, and designers into the development process has been shown to improve usability and effectiveness, particularly in the context of complex healthcare environments [16, 18].

3 Methodology

3.1 Workshop Design and Materials

We conducted an interdisciplinary workshop that brought together a diverse group of participants, including HCI professionals and clinical experts specializing in DLB. The goal was to generate ideas for leveraging smartphone and smartwatch sensors to gather multidimensional data related to DLB symptoms and to collaboratively address challenges in monitoring DLB symptoms and propose novel ways to leverage digital tools for data collection.

To ensure all participants, regardless of their disciplinary background, had a shared understanding of the problem space, the workshop began with a comprehensive presentation. This session covered key clinical aspects of DLB, including symptoms and current diagnostic practices; the opportunities and challenges associated

with integrating digital sensors into DLB diagnosis and monitoring; a detailed review of the challenges identified through prior research; and the overarching aims and structure of the workshop, along with instructions for participatory activities. To complement the presentation and provide extended resources for discussion, participants were given detailed handouts. These materials encompassed definitions and subcategories of DLB symptoms, along with clinically validated assessment tools based on established guidelines [15], and a summary of smartphone and smartwatch sensor functions, highlighting their applications in dementia research. The handouts also presented key challenges in DLB monitoring, as identified in the literature, to guide participants during the brainstorming sessions. These challenges included addressing the variability and heterogeneity of DLB symptoms across patients, capturing symptom fluctuations through longitudinal and continuous data collection, ensuring the acquisition of multimodal data to reflect the disease’s multidimensional nature, and accounting for covariates and confounding factors. These considerations were emphasized as the workshop aimed to explore innovative ways to redesign digital tools to address these critical gaps. Additionally, preprinted cards representing key symptom categories and the capabilities of smartphone and smartwatch sensors, along with sticky notes, were provided to facilitate interactive discussions and ideation during the workshop.

3.2 Participants

We recruited participants for the workshop using convenience sampling within a clinical research consortium working on DLB and a local Human-Computer Interaction research group. In total, we recruited 11 participants, 3 clinicians, 4 Human-Computer Interaction researchers, and 4 researchers with medical computer science backgrounds. The participants spanned all career levels, with two medical doctors, three HCI professors, two postdocs, two PhD students, and two master’s students.

3.3 Workshop Structure

The workshop featured structured activities designed to guide participants through the process of rethinking digital data collection methods for DLB. After the presentation and distribution of handouts, participants were divided into two interdisciplinary groups, each including at least one clinical expert in DLB to provide domain-specific insights. Participants were instructed to utilize the provided materials, such as symptom and sensor cards, and sticky notes, to brainstorm innovative solutions addressing the identified challenges. Specifically, they were tasked with designing approaches that:

- Capture the variability and heterogeneity of DLB symptoms across all patients with different profiles.
- Account for symptom fluctuations through longitudinal and continuous monitoring.
- Address the multidimensional nature of DLB by integrating data from multiple domains
- Minimize intrusiveness while considering the effects of covariates and confounding factors.

The groups were encouraged to think about how smartphone and smartwatch sensors could be utilized to achieve these goals

and to propose novel tasks or methods for integrating passive and active data collection. Each group worked collaboratively, using the preprinted cards to map symptoms to sensor capabilities and brainstorm potential scenarios for data collection. Sticky notes were used to annotate ideas, group concepts, and outline their proposed solutions. Once the group discussions concluded, participants presented their ideas and solutions in a plenary session. This allowed for cross-group dialogue, where ideas were refined, integrated, and aligned with the overarching goals of the workshop. Detailed notes from the discussions were collected to capture all discussed insights. A photo documenting the note boards from the discussions is presented in Figure 1.

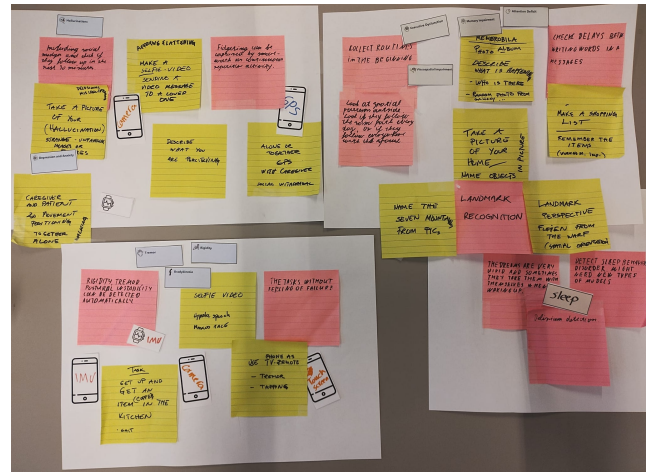


Figure 1: Notes and collaborative boards from the workshop discussions.

3.4 Data Collection and Analysis

Insights from the workshop were captured through detailed note-taking by facilitators and group members and included sketches, scenarios and other artifacts that were produced during the workshop. The analysis followed this process:

- (1) Initial Review: Workshop notes were reviewed to identify key topics, themes, and recurring patterns.
- (2) Theme Development: Gathered data were grouped into broader themes that aligned with the challenges and goals of the workshop.
- (3) Idea Generation and Refinement: Building on the developed themes, a preliminary framework was developed that integrated all the ideas into a cohesive, holistic approach aligned with the workshop objectives. Following this, a series of follow-up consultations were held with selected experts who had participated in the workshop, to evaluate and further refine the proposed solutions.

4 Findings and Insights

We have grouped the synthesized insights into two overarching themes: Task Design insights (T1–T5) and Sensing insights (S1–S5).

Task Design insights focus on findings from discussions about designing tasks that facilitate holistic data collection while seamlessly integrating into daily life, minimizing burden, and maximizing engagement. Sensing insights explore approaches for utilizing various sensing modalities to capture diverse, multimodal symptom-related data, leveraging tools such as smartphone cameras, touchscreens, and wearables to enable holistic and continuous monitoring.

We present further details of individual subthemes below.

4.1 Task design insights

• T1: Leveraging Data from Daily Life

A key theme from the workshop was the importance of grounding data collection in patients' existing daily activities. This approach minimizes the burden on patients by integrating tasks into routines they already find familiar and manageable. Examples included analyzing text-messaging patterns, tracking phone usage, and using photo albums to design cognitive tasks like scene descriptions or recognizing familiar faces. This unobtrusive approach aligns with patients' existing habits and can provide meaningful insights with minimal burden.

• T2: Alternative and Overlapping Tasks

Recognizing the variability in patient symptoms, preferences, and engagement levels, participants emphasized the need for multiple alternative tasks that provide overlapping data. For example, cognitive data could be assessed through both video recordings and active text conversations, ensuring that necessary data is captured even if one task type is unsuitable for a patient. This redundancy improves data reliability and inclusivity across a diverse patient population, and when all types are collected can be further used for triangulation.

• T3: Personalized and Familiar Task Design

Personalization was identified as essential for ensuring task relevance and engagement. Tasks could incorporate familiar objects, environments, or routines, such as identifying items in a patient's home or landmarks of the city they live.

• T4: Minimizing Perceived Failures

One clinical expert with extensive experience with patients stressed the importance of fostering a sense of achievement in task design. Tasks should be adaptable to patients' abilities, with supportive feedback mechanisms that prevent frustration or feelings of inadequacy.

• T5: Augmenting Diagnostic Tasks with Care Benefits

Participants suggested designing tasks that not only collect data but also promote physical or cognitive well-being. For example, a gait-assessment task could double as a light exercise routine, or memory tasks could serve as cognitive stimulation exercises.

4.2 Sensing insights

• S1: Combining Multiple Data Types for Holistic Insights

Another theme was the integration of diverse data types, such as motion, cognitive, and neuropsychiatric metrics, within a single task. For instance, guiding patients to retrieve

objects could simultaneously capture gait data, balance stability, and cognitive decision-making. This approach ensures that the interplay between different symptoms is considered, fostering a more comprehensive understanding of patient health.

• S2: Balancing Active and Passive Data Collection

To address usability and compliance, the importance of designing active tasks with the aim of being complementary to passive data collection was highlighted. Active tasks, such as asking patients to record a video or complete a specific activity, provide targeted insights. Meanwhile, passive data from wearable sensors while performing that activity (e.g., gait asymmetry, heart rate, or balance measures) ensures a complementary, unobtrusive layer of monitoring.

• S3: Camera—the Central Tool for Data Collection

Discussions identified that a smartphone camera has the potential to be a central data-gathering tool due to its accessibility and versatility. The camera is intuitive for users and can capture a broad range of data types, including facial expressions, speech patterns, and motor functions. Example tasks include:

- Recording Video Selfies: Patients are instructed to record short video clips of themselves speaking or performing simple tasks, providing data on cognitive, neuropsychiatric, and motor symptoms.
- Object Identification Tasks: Patients are instructed to identify or describe items in photos or their immediate environment, offering insights into cognitive and verbal abilities.
- Photographing Visual Hallucinations: Instructing patients to photograph perceived anomalies, enabling precise documentation and validation with object detection algorithms for reassurance and analysis.

• S4: Social Interaction Monitoring and Promotion

Social interaction was identified as a rich source of multimodal data, covering cognitive, neuropsychiatric, speech, and motor elements. Monitoring interaction patterns—such as call duration, frequency of communication, and writing patterns in messages—can provide insights into multiple aspects of a patient's health. Additionally, promoting social engagement has benefits for physical and mental well-being, making it a dual-purpose focus. Tasks designed to encourage social interactions, such as recording messages for family members, could simultaneously enhance quality of life and data collection.

• S5: Dream Residual and Emotional State Tracking

Some ideas included prompting patients to describe residual dream states after waking up and leveraging wearable devices to detect sleep-wake transitions. This approach could provide unique insights into multiple symptoms, such as patients' cognitive state and sleep-related symptoms.

5 AI-Driven Personal Assistant for Holistic and Personalized Data Collection

Building on the workshop discussions and findings, the digital assistant concept emerged as a unifying framework to integrate the

discussed ideas. We propose an AI-powered digital assistant for collecting comprehensive and multimodal data to assess, monitor, and support patients with DLB. The assistant's primary objective is to gather a holistic and dynamic patient profile by integrating passive and active data collection methods, supporting both research and care-oriented goals.

The digital assistant will serve as a central hub for personalized data collection, mediating daily interactions between the user and the system. We design active tasks with a focus on integration into users' daily routines. Tasks are aligned with regular activities, such as taking photos, recording short video messages, or responding to simple prompts, thus minimizing disruption. (T1) Based on workshop discussions, the camera is prioritized as the primary data acquisition tool due to its accessibility, ease of use, and rich data output. (S3) We also include other tasks employing the same principle of aligning with daily activities, ensuring coverage of all necessary symptom categories despite individual variability in symptoms, situations, and preferences. (T1, T2) The assistant will deliver pre-defined, context-sensitive instructions. For example, users may be prompted to capture an image when experiencing an unexpected visual perception (potentially a hallucination), which is then analyzed using object detection algorithms and compared with the user's report to distinguish true hallucinations from misperceptions. This approach provides both monitoring/assessment capabilities and reassurance for patients. (T5) Interaction modalities include voice, video messages, and text. Figure 2 provides a schematic representation of the concept.

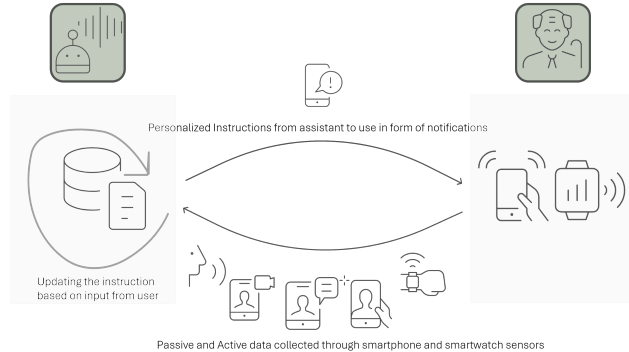


Figure 2: Schematic framework of the AI-powered assistant for personalized DLB data collection, showing bidirectional communication and adaptive task management

A key innovation of this idea is the dynamic adaptation of daily tasks based on user feedback and collected data. Given the diverse range of symptoms in DLB, there are numerous possible active tasks for data collection. However, not all patients experience the same combination of symptoms, nor do they engage with digital tools in the same way. Daily routines, interaction preferences (e.g. voice versus text), and the type of passive data generated vary between individuals. This variability underscores the importance of a personalized approach, making the AI-driven assistant's ability to adapt tasks a useful feature for ensuring relevance and maximizing engagement. Rather than overwhelming patients with a fixed set

of tasks or relying on them to initiate engagement, the assistant dynamically adapts task selection based on each individual's profile, symptoms, and interaction preferences. By administering only the most relevant tasks, the assistant improves compliance, ensures data collection remains both targeted and feasible, and ultimately improves the quality and richness of the gathered data. (T3, T4)

Personalization is achieved through several mechanisms. Firstly, tasks are dynamically adjusted based on data completeness; the assistant analyzes previously gathered data (creating a feedback loop that informs subsequent task selection) and prioritizes tasks to address information gaps, reducing redundancy. Secondly, task selection is informed by disease progression, the level of patient involvement, and individual abilities and preferences. While the digital assistant may provide general recommendations (e.g., using video messages), it will prioritize personalized instructions to optimize task relevance, minimize intrusiveness, and maximize user engagement. This adaptive strategy, combined with the design of the overlapping tasks for each target feature, ensures comprehensive data capture even if specific active tasks are not completed due to user limitations or lack of interest. (S1, S2) The integration of passive data streams with multimodal active data further enriches the dataset, facilitating the search for novel digital biomarkers.

Aligned with another key theme of the workshop, the design of active tasks also incorporates secondary objectives, such as promoting social engagement through conversational prompts and encouraging daily exercise. (T5, S4)

6 Discussion and Future Directions

This paper presents the first phase of a research project aimed at developing improved data collection methodologies for individuals with DLB, addressing limitations inherent in current clinical and digital approaches. This work sought to establish a foundational framework for leveraging the opportunities presented by smartphone and wearable sensor technologies to address challenges in both traditional clinical assessments and existing digital data-gathering methods. This requires close collaboration between experts in the clinical domain and HCI, and therefore, this initial phase focused on foundational design principles, bringing together expertise from both fields. These collaborative discussions laid the groundwork for the digital assistant concept and personalized, context-aware data acquisition principles.

While this phase focused on establishing the core conceptual framework with expert input, the next steps involve participatory co-design workshops with all relevant stakeholders, especially end-users. These workshops will bring together patients, caregivers, and professionals with a broader range of expertise, including designers, data scientists, clinicians, and nurses, to collaboratively refine the designed tasks and application features. This participatory approach is essential for ensuring the final system's usability, acceptability, and effectiveness. By directly involving end-users in the design process, we aim to create a system that is not only scientifically rigorous but also integrated into the daily lives of individuals with DLB and their caregivers. Future work will also address the technical implementation of the assistant, including the development of algorithms for data analysis, task adaptation, and

user interaction. Furthermore, we plan to evaluate the system's usability, feasibility, and impact on data quality and patient outcomes in a set of user studies.

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